

Note 12.1 Three-Dimensional Coordinate Systems

Basic Concepts

- right-handed coordinate frame
- rectangular coordinates(Cartesian coordinates for space)
- coordinate planes
- octants(generally use the first octant)

Distance and Spheres in Space

The Distance Between $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$

$$|P_1P_2| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

The Standard Equation for the Sphere of Radius a and Center (x_0, y_0, z_0)

$$(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = a^2$$

and we can use the inequalities to judge a point inside or outside a sphere.